QML Series #49

Hybrid Quantum-Classical Computing

Combining the Power of Classical AI & Quantum Computing

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Hybrid Quantum-Classical Computing combines the strengths of classical computers and quantum computers, allowing them to work together to solve machine learning problems efficiently.

LEARNING ROADMAP

Classical AI
↓
Feature Extraction
↓
Quantum Computing
↓
Hybrid Quantum-Classical Model
↓
Prediction

1. WHY HYBRID COMPUTING?

Today's quantum computers are not yet powerful enough to solve large AI problems on their own.

Current quantum computers have:

- ✗ Limited qubits
- ✗ Quantum noise
- ✗ Decoherence
- ✗ Short circuit depth
- ✗ Measurement errors

These systems belong to the NISQ Era (Noisy Intermediate-Scale Quantum).

Solution

Instead of replacing classical computers, we combine both.

Classical Computer
+
Quantum Computer
↓
Hybrid Computing

2. WHAT IS A HYBRID QUANTUM-CLASSICAL MODEL?

A Hybrid Quantum-Classical Model is a machine learning system where:

- Classical Computer performs part of the computation
- Quantum Computer performs another part
- Both work together during training and prediction

Simple Definition

Hybrid Model
=
Classical AI
+
Quantum Computing

3. WHY NOT USE ONLY QUANTUM COMPUTERS?

Suppose your input is Brain MRI
224 × 224 × 3

Total pixels
150,528 Features

Processing every pixel directly would require far more qubits than today's hardware provides.

Current quantum computers cannot efficiently process such large raw inputs.

4. THE HYBRID SOLUTION

Brain MRI
CNN / Vision Transformer
128 Features
Quantum Circuit
Measurement
Dense Layer
Prediction

This is the most common architecture in modern Quantum Machine Learning.

COMPLETE HYBRID WORKFLOW

Classical Data
Feature Extraction
Quantum Data Encoding
Parameterized Quantum Circuit
Measurement
Classical Optimizer
Update Parameters
Repeat

The optimizer still runs on the classical computer.

6. COMPONENTS OF A HYBRID MODEL

Step 1 – Classical Input

- MRI Images
- CT Scans
- Tabular Data
- Text
- Audio

Step 2 – Classical Feature Extraction

Popular Models

- CNN
- EfficientNet
- ResNet
- Vision Transformer (ViT)
- Dense Networks

Output: 128 Feature Vector

Step 3 – Quantum Data Encoding

- Classical features are converted into quantum states.

Common Methods

- Angle Embedding
- Amplitude Embedding
- Basis Embedding

Features
Qubit Rotations

Step 4 – Parameterized Quantum Circuit (PQC)

Angle Embedding
RY(θ)
RX(θ)
CNOT
RZ(θ)

Trainable parameters θ are optimized during learning.

Step 5 – Measurement

Measure expectation values.
↓
Example: Pauli-Z
↓
0.73 | -0.41 | 0.88 | ...

Quantum information becomes classical values.

Step 6 – Classical Prediction Layer

Quantum Output
↓
Dense Layer
↓
Softmax
↓
Prediction

7. COMPLETE HYBRID ARCHITECTURE

Input Image
EfficientNet-B0
Feature Vector
Angle Embedding
Parameterized Quantum Circuit
Expectation Values
Fully Connected Layer
Prediction

This architecture appears in many modern Hybrid QML research papers.

8. MEDICAL IMAGE CLASSIFICATION EXAMPLE

512 Features
↓
PCA / Linear Layer
↓
16 Features
↓
4 Features
↓
4 Qubits
Quantum Circuit Layers
Measurement
Classifier
↓
Prediction

9. WHY REDUCE FEATURES?

Without Reduction

512 Features
↓
512 Qubits
↓
✗ Impossible Today

With Reduction

512 Features
↓
16 Features
↓
4 Features
↓
4 Qubits
✓ Practical

TRAINING FLOW

Image
CNN
Quantum Layer
Prediction
Loss
Classical Optimizer
Update θ
Repeat

Only the trainable quantum parameters (θ) are updated during optimization.

11. HYBRID NEURAL NETWORK STRUCTURE

Input Features
↓
Linear Layer
↓
Quantum Layer (PQC)
↓
Linear Layer
↓
Prediction

This is one of the simplest Hybrid QML architectures.

12. CLASSICAL vs HYBRID vs FULLY QUANTUM

Aspect	Classical AI	Hybrid QML	Fully Quantum
Computing	CPU / GPU Only	CPU + Quantum Processor	Quantum Processor Only
Technology	Mature	Most Practical Today	Future Goal
Handles Data	Large Scale	Limited by Qubit Count	Limited by Current Hardware
Quantum Advantage	No	Potential Quantum Advantage	Full Quantum Processing

WHERE ARE HYBRID MODELS USED?

- ✓ Image Classification
- ✓ Medical Imaging
- ✓ Drug Discovery
- ✓ Materials Science
- ✓ Quantum Chemistry
- ✓ Financial Prediction
- ✓ NLP Research

WHY HYBRID MODELS DOMINATE RESEARCH

Classical AI (Heavy Computation)
↓
Quantum Circuit (Compact Feature Learning)
↓
Classical Optimizer (Parameter Updates)

- ✗ Limited number of qubits
- ✗ Quantum hardware is noisy
- ✗ Data encoding can become a bottleneck
- ✗ Quantum circuit execution is slower than classical layers
- ✗ Training often requires many quantum circuit evaluations

This combination is the most practical approach for today's Quantum Machine Learning.

CLASSICAL vs QUANTUM RESPONSIBILITIES

Classical Computer (What It Does Best)

- Data preprocessing
- Feature extraction (CNN / ViT)
- Loss calculation
- Parameter optimization
- Final prediction

↔

Quantum Computer (What It Does Best)

- Quantum state preparation
- Parameterized Quantum Circuit
- Superposition & Entanglement
- Quantum Feature Learning
- Expectation Value Measurement

INTERVIEW QUESTIONS

Q1. What is a Hybrid Quantum-Classical Model?
A: A machine learning model where classical computers and quantum computers work together during training and inference.

Q2. Why are Hybrid Models preferred today?
A: Because current quantum computers have limited qubits and high noise, while classical computers efficiently perform preprocessing, optimization, and large-scale computation.

Q3. Which part usually runs on the Quantum Computer?
A: The Parameterized Quantum Circuit (Quantum Layer).

Q4. Which part usually runs on the the Classical Computer?
A: Feature extraction, data preprocessing, parameter optimization, final prediction layers, ...

Q5. Why is Feature Reduction important?
A: Because today's quantum processors [illegible] of features directly due to number of qubits.

QUICK SUMMARY

- ✓ Hybrid models combine the strengths of Classical AI and Quantum Computing.
- ✓ Classical networks extract meaningful features from large datasets.
- ✓ Quantum circuits process compact features using PQCs.
- ✓ Measurements convert quantum outputs to classical values.
- ✓ A classical optimizer updates the trainable quantum parameters.
- ✓ Hybrid models balance accuracy, trainability, and hardware limitations.

Hybrid Quantum-Classical Computing is the bridge between today's powerful classical AI and tomorrow's fault-tolerant quantum computers. It is the foundation of almost all practical Quantum Machine Learning systems in the NISQ era.

QML Series #50

Variational Quantum Algorithms (VQAs)

HARIKARAN M
AI Researcher

The Engine Behind Modern Quantum Machine Learning

A Variational Quantum Algorithm (VQA) is a hybrid algorithm where classical and quantum computers work together in a loop to solve optimization and machine learning problems.

MODULE ROADMAP

Classical Optimization
↓
Parameterized Quantum Circuit (PQC)
↓
Variational Quantum Algorithm (VQA)
↓
VQE • QAOA • VQC • QNN
↓
Hybrid Quantum AI Applications

1. What is a Variational Quantum Algorithm (VQA)?

Simple Definition

Classical Computer	Quantum Computer
Updates trainable parameters (θ)	Executes a PQC
Computes loss/cost	Measures the quantum state
Runs the optimizer	Returns the output

Formula

Quantum Circuit + Classical Optimizer = Variational Quantum Algorithm (VQA)

Real-Life Analogy

Throw Ball → Miss Basket → Coach Gives Feedback → Adjust Technique → Throw Again

Repeat until you score.

Why Do We Need VQAs?

Current quantum computers are NISQ devices:

- ✗ Limited qubits
- ✗ Noise
- ✗ Decoherence
- ✗ Short circuit depth
- ✗ Measurement errors

Instead of relying only on quantum hardware:

Classical Computer
↓
Optimization
↓
Quantum Computer

Both work together.

The Variational Optimization Loop

1. Initialize Parameters (θ)
2. Encode Data
3. Parameterized Quantum Circuit
4. Measurement
5. Compute Loss
6. Classical Optimizer
7. Update Parameters (θ)
8. Repeat Until Convergence

This repeated process is called the Variational Optimization Loop.

Six Building Blocks of a VQA

1 Classical Input

- Image
- Tabular Data
- Text
- Audio

→

2 Data Encoding

Convert classical data into qubits.

- Angle Embedding
- Amplitude Embedding
- Basis Embedding

Features
↓
Qubit Rotations

→

3 Parameterized Quantum Circuit

Trainable quantum gates

$RY(\theta_1)$
$RX(\theta_2)$
CNOT
$RZ(\theta_3)$

→

4 Measurement

Measure expectation values.

Example

Pauli-Z
↓
0.73 | -0.41 | 0.88 | ···

→

5 Cost (Loss) Function

Examples

- Cross Entropy
- Mean Squared Error
- Energy Expectation

→

6 Classical Optimizer

Examples

- Adam
- COBYLA
- SPSA
- Gradient Descent

Updates θ after every iteration.

Complete VQA Architecture

Classical Data
↓
Data Encoding
↓
Parameterized Quantum Circuit
↓
Measurement
↓
Loss / Cost Function
↓
Classical Optimizer
↓
Update Parameters (θ)
↓
Repeat

The optimizer runs on the classical computer.

Four Major Variational Quantum Algorithms

Algorithm	Purpose	Common Applications
VQE	Ground-state energy estimation	Quantum Chemistry, Drug Discovery
QAOA	Optimization	Routing, Finance, Scheduling
VQC	Classification	Medical Imaging, AI
QNN	Machine Learning	Pattern Recognition, Hybrid AI

1 VQE (Variational Quantum Eigensolver)

Hamiltonian
↓
Parameterized Circuit
↓
Energy Measurement
↓
Optimizer
↓
Minimum Energy

Applications

- Quantum Chemistry
- Drug Discovery
- Molecular Simulation
- Materials Science

2 QAOA (Quantum Approximate Optimization Algorithm)

Optimization Problem
↓
Quantum Circuit
↓
Measurement
↓
Optimizer
↓
Best Solution

Applications

- Vehicle Routing
- Portfolio Optimization
- Scheduling
- Max-Cut Problems

3 VQC (Variational Quantum Classifier)

Image
↓
Feature Extraction
↓
Quantum Circuit
↓
Prediction

Applications

- Brain MRI Classification
- Disease Detection
- Image Recognition

4 QNN (Quantum Neural Network)

Input
↓
Quantum Layer (PQC)
↓
Measurement
↓
Output

A quantum version of a neural network using parameterized quantum circuits instead of hidden neurons.

Medical Image Classification Example

- MRI Image
- EfficientNet / CNN
- Feature Reduction
- Angle Embedding
- Parameterized Quantum Circuit
- Expectation Values
- Dense Layer
- Prediction

Training Loop

Prediction
↓
CrossEntropy Loss
↓
Adam Optimizer
↓
Update Quantum Parameters (θ)

This complete pipeline is a Variational Quantum Algorithm.

Classical NN vs VQA

Classical Neural Network	Variational Quantum Algorithm
Neural Network	Parameterized Quantum Circuit
Weights (W)	Quantum Parameters (θ)
Forward Pass	Quantum Circuit Execution
Backpropagation	Classical Optimization + Quantum Gradients
Loss Function	Cost Function
Adam / SGD	Adam, COBYLA, SPSA

Classical vs Hybrid vs Fully Quantum

Aspect	Classical AI	Hybrid QML	Fully Quantum
Compute	CPU / GPU Only	CPU + Quantum Processor	Quantum Processor Only
Maturity	Mature Technology	Most Practical Today	Future Goal
Handles Large Data	Yes	Yes (via classical)	Limited by Hardware
Quantum Advantage	No	Potential Quantum Advantage	Full Quantum Processing

Where Are VQAs Used?

AI & Machine Learning	QNN, VQC
Medical Imaging	Tumor Classification
Drug Discovery	Molecular Energy
Quantum Chemistry	VQE
Finance	Portfolio Optimization
Logistics	QAOA Routing
Materials Science	Ground-State Simulation

Why Hybrid Models Dominate Research?

Current quantum computers are still in the NISQ Era. Therefore,

Classical AI
↓ Heavy Computation
Quantum Circuit
↓ Compact Feature Learning
Classical Optimizer
Parameter Updates

Advantages

- ✓ Works on today's NISQ hardware
- ✓ Hybrid quantum-classical learning
- ✓ Requires relatively few qubits
- ✓ Integrates with PennyLane, Qiskit, PyTorch & TensorFlow
- ✓ Foundation of modern Quantum Machine Learning

Limitations

- ✗ Requires many quantum circuit evaluations
- ✗ Sensitive to hardware noise
- ✗ Can suffer from barren plateaus
- ✗ Limited by today's qubit count
- ✗ Quantum circuit execution is slower than classical layers

Classical vs Quantum Responsibilities

Classical Computer	Quantum Computer
Data preprocessing	Quantum state preparation
Feature extraction (CNN / ViT)	Parameterized Quantum Circuit
Loss calculation	Superposition & Entanglement
Parameter optimization	Expectation Value Measurement
Final prediction	

Interview Questions

1. What is a Variational Quantum Algorithm (VQA)?
2. Why are VQAs important?
3. Name four popular VQAs.
4. What does the quantum computer do?
5. What does the classical computer do?
6. Why is Feature Reduction important?

Key Takeaways

- ✓ A VQA combines a Parameterized Quantum Circuit with a Classical Optimizer.
- ✓ Training follows the loop: Initialize → Execute → Measure → Compute Loss → Update → Repeat
- ✓ Four important VQAs: VQE (Energy) • QAOA (Optimization) • VQC (Classification) • QNN (ML)
- ✓ VQAs are the foundation of modern Quantum Machine Learning in the NISQ era.

Easy Memory Trick

Classical AI + Large Data Handling + Quantum Computing + Complex Feature Learning = Smarter AI

VQAs are the bridge between today's powerful classical AI and tomorrow's fault-tolerant quantum computers.
They make practical Quantum Machine Learning possible in the NISQ era.

QML Series #51

Classical Optimizers for Quantum Circuits

HARIKARAN M
AI Researcher

The Learning Engine of Quantum Machine Learning

A Classical Optimizer updates the trainable parameters (θ) of a quantum circuit to minimize the loss (cost) function and improve predictions through iterative learning.

MODULE ROADMAP

Quantum Circuit
↓
Measurement
↓
Loss Function
↓
Classical Optimizer
↓
Update Parameters (θ)
↓
Better Quantum Model

1. WHAT IS A CLASSICAL OPTIMIZER?

Simple Definition

A Classical Optimizer is an algorithm that updates the trainable parameters (θ) of a Parameterized Quantum Circuit (PQC) to minimize the loss (cost) function.

Think of it like this:

Optimizer = Teacher of the Quantum Circuit

Quantum Computer
- ✓ Executes the circuit
- ✓ Measures the state
- ✓ Returns the output

Classical Optimizer
- ✓ Updates parameters (θ)
- ✓ Computes loss/cost
- ✓ Runs the optimizer

HOW DOES LEARNING HAPPEN?

Before Training

$RY(\theta_1)$, $RX(\theta_2)$, $RZ(\theta_3)$

$\theta_1 = 0.12$
$\theta_2 = 0.55$
$\theta_3 = -0.30$

Loss = 0.82

After Optimization

$RY(\theta_1)$, $RX(\theta_2)$, $RZ(\theta_3)$

$\theta_1 = 0.21$
$\theta_2 = 0.48$
$\theta_3 = -0.17$

Loss = 0.54

The optimizer updates θ to improve predictions.

OPTIMIZATION LOOP

1. Initialize Parameters (θ)
2. Encode Data
3. Run Quantum Circuit
4. Measure Output
5. Compute Loss
6. Classical Optimizer
7. Update Parameters (θ)
8. Repeat Until Convergence

This loop powers every Variational Quantum Algorithm (VQA).

4. WHY DO WE NEED AN OPTIMIZER?

Without Optimizer

Parameters
↓
Never Change
↓
Same Prediction Forever
✗

With Optimizer

Parameters
↓
Updated Every Epoch
↓
Better Predictions
✓

5. TYPES OF CLASSICAL OPTIMIZERS

Classical Optimizers

Gradient-Based (Uses Gradients)
- Adam ★
- SGD
- RMSProp
- Gradient Descent

Gradient-Free (No Gradients)
- COBYLA ★
- SPSA ★
- Nelder-Mead
- Powell

6. SIX BUILDING BLOCKS OF A VQA

1. **Classical Input**
   - Image
   - Tabular
   - Text
   - Audio
2. **Data Encoding**
   - Angle Embedding
   - Amplitude Embedding
   - Basis Embedding
3. **Parameterized Quantum Circuit** — Trainable Quantum Gates
4. **Measurement** — Measure Expectation Values
5. **Cost (Loss) Function**
   - Cross Entropy
   - MSE
   - Energy Expectation
6. **Classical Optimizer**
   - Adam
   - COBYLA
   - SPSA
   - GD...

Updates θ after every iteration.

7. GRADIENT-BASED OPTIMIZERS (Use Gradients)

① Gradient Descent (GD)

Formula

$$\theta_{new} = \theta_{old} - \eta \frac{\partial L}{\partial \theta}$$

Best For: Learning basics

Pros
- ✓ Simple
- ✓ Easy to understand

Cons
- ✗ Slow convergence
- ✗ Sensitive to learning rate

② SGD (Stochastic GD)

Uses mini-batch instead of full dataset.

Mini Batch
↓
Gradient
↓
Parameter Update

Pros
- ✓ Faster than GD
- ✓ Scales well

Cons
- ✗ Noisy updates
- ✗ May oscillate

Adam (Most Popular in QML) ★

Combines momentum + adaptive learning rate

Gradient → Momentum
Adaptive LR → Update θ

Advantages
- ✓ Fast convergence
- ✓ Stable training
- ✓ Excellent for Hybrid QML
- ✓ Works with PennyLane & PyTorch

RMSProp

Adjusts learning rate using recent gradients.

Advantages
- ✓ Stable
- ✓ Handles noisy optimization

Limitation

Less common than Adam in QML.

8. GRADIENT-FREE OPTIMIZERS (No Gradients)

★ COBYLA (Constrained Optimization BY Linear Approximation)

Parameters
↓
Evaluate Loss
↓
Estimate Better Direction
↓
Update Parameters
↓

Best For
- ✓ VQE, QAOA
- ✓ Noisy optimization

Nelder-Mead

Uses a geometric simplex to search for better parameters.

Good for

Small optimization problems

Powell

Optimizes one direction at a time.

Useful when

gradients are unavailable.

★ SPSA (Simultaneous Perturbation Stochastic Approximation)

Requires only two circuit evaluations regardless of parameter size.

θ
↓
θ + Δ | θ − Δ
↓
Estimate Gradient
↓
Update θ

Advantages
- ✓ Very few circuit evaluations
- ✓ Noise tolerant
- ✓ Excellent for IBM Quantum

9. PENNYLANE OPTIMIZERS

- ✓ AdamOptimizer
- ✓ GradientDescentOptimizer
- ✓ RMSPropOptimizer
- ✓ MomentumOptimizer
- ✓ AdagradOptimizer
- ✓ NesterovMomentumOptimizer

10. QISKIT OPTIMIZERS

- ✓ COBYLA
- ✓ SPSA
- ✓ ADAM
- ✓ L-BFGS-B
- ✓ GradientDescent
- ✓ Nelder-Mead

11. WHICH OPTIMIZER SHOULD YOU CHOOSE?

Scenario	Recommended Optimizer
PennyLane + PyTorch	★ Adam
Hybrid Quantum ML	★ Adam
Quantum Simulator	★ Adam
IBM Quantum Hardware	★ SPSA
VQE	★ COBYLA / L-BFGS-B
QAOA	★ COBYLA / SPSA
Noisy Optimization	★ SPSA
No Gradient Available	★ COBYLA

12. MEDICAL IMAGE CLASSIFICATION EXAMPLE

MRI Image
EfficientNet / CNN
Feature Reduction
Angle Embedding
Parameterized Quantum Circuit
Expectation Values
Dense Layer
Prediction

Training Loop

Prediction
↓
CrossEntropy Loss
↓
Adam Optimizer
↓
Update Quantum Parameters (θ)

This is the most common Hybrid QML training pipeline.

13. CLASSICAL vs VQA (TRAINING)

	Classical Neural Network	Variational Quantum Algorithm
Model	Neural Network	Parameterized Quantum Circuit
Parameters	Weights (W)	Quantum Parameters (θ)
Forward Pass	Forward Pass	Quantum Circuit Execution
Update Method	Backpropagation	Classical Optimization + Quantum Gradients
Loss	Loss Function	Cost Function
Optimizers	Adam / SGD	Adam, COBYLA, SPSA

14. OPTIMIZER COMPARISON

Optimizer	Gradient Needed	Noise Tolerance	Best Use
Gradient Descent	✓	★	Learning Basics
SGD	✓	★★	Mini-Batch Training
RMSProp	✓	★★★	Adaptive Optimization
Adam	✓	★★★	Hybrid QML & Simulators
COBYLA	✗	★★★	VQE, QAOA
Nelder-Mead	✗	★★	Small Problems
Powell	✗	★★★	Small Problems
SPSA	Approx	★★★★★	Real Quantum Hardware

15. ADVANTAGES

- Works on today's NISQ hardware
- Hybrid quantum-classical learning
- Flexible for AI, chemistry & optimization
- Compatible with PennyLane, Qiskit, PyTorch & TensorFlow
- Foundation of modern Quantum Machine Learning

16. LIMITATIONS

- ✗ Requires many quantum circuit evaluations
- ✗ Sensitive to hardware noise
- ✗ Can suffer from barren plateaus
- ✗ Limited by today's qubit count

17. INTERVIEW QUESTIONS

1. What is a Classical Optimizer?
2. Which optimizer is most common in PennyLane?
3. Why is SPSA popular on real quantum hardware?
4. When should you use COBYLA?
5. Gradient-Based vs Gradient-Free?

Gradient-Based (Uses gradients)

Adam, SGD, RMSProp, GD, ...

VS

Gradient-Free (No gradients)

COBYLA, SPSA, Powell, ...

18. KEY TAKEAWAYS

- A Classical Optimizer teaches a quantum circuit by updating its parameters.
- Every VQA follows: Initialize → Execute → Measure → Compute Loss → Optimize → Repeat
- Adam is preferred for simulators; SPSA is best for real hardware.
- COBYLA is widely used in VQE and QAOA when gradients are unavailable.

ONE-LINE SUMMARY

A Variational Quantum Algorithm (VQA) is a hybrid optimization framework where a parameterized quantum circuit learns through repeated interaction with a classical optimizer, enabling practical Quantum Machine Learning on today's NISQ quantum computers.

Series #52

Become a Quantum Machine Learning (QML) Engineer

Quantum Machine Learning Frameworks

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AI Researcher

Quantum Machine Learning frameworks connect classical AI libraries with quantum hardware and simulators, making it easier to build, train, and deploy hybrid quantum models.

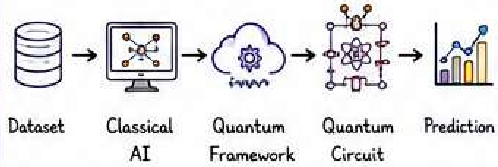
1 Learning Roadmap

Classical ML
↓
Quantum Computing
↓
Quantum Frameworks
↓
Hybrid QML Models
↓
Real Applications

2 What is a QML Framework?

A Quantum Machine Learning framework connects classical AI libraries with quantum hardware and simulators, making it easier to build, train, and deploy hybrid quantum models.

How It Works



It acts as a bridge between classical computers and quantum systems.

3 Popular Frameworks

1. PennyLane
- Best for Hybrid QML
- Works with PyTorch & TensorFlow
- Automatic gradients
- Beginner friendly

2. Qiskit
- IBM Quantum ecosystem
- Circuit design
- Real quantum hardware
- Research focused

3. TensorFlow Quantum (TFQ)
- TensorFlow integration
- Quantum neural networks
- Hybrid deep learning

4. Cirq
- Google's quantum SDK
- Circuit simulation
- NISQ research

4 Framework Comparison

Framework	Best Use	Beginner	Hardware
PennyLane	Hybrid QML	★★★★★	IBM, IonQ, Simulators
Qiskit	Quantum Algorithms	★★★★☆	IBM Quantum
TensorFlow Quantum (TFQ)	Deep Learning	★★★☆	TensorFlow Ecosystem
Cirq	Research	★★★☆	Google Simulators

5 Typical Hybrid Workflow

Image / Data
↓
CNN / Feature Extractor
↓
Quantum Framework
↓
Parameterized Quantum Circuit
↓
Measurement
↓
Prediction

6 Why Frameworks Matter?

- Simplify quantum programming
- Connect AI with quantum circuits
- Automatic gradient computation
- Run on simulators and real hardware

7 Best Choice Guide

Beginner →	PennyLane
IBM Quantum Users →	Qiskit
TensorFlow Users →	TFQ
Research Circuits →	Cirq

8 Key Takeaways

- Frameworks simplify QML development.
- PennyLane is the most popular choice for Hybrid QML.
- Qiskit is ideal for IBM Quantum hardware.
- Hybrid frameworks connect classical AI with quantum circuits.
- They enable building practical QML models on today's NISQ devices.

Bonus: Hybrid Strength

Classical AI		Quantum Computing
Handles large data	+	Superposition
Feature extraction		Entanglement
Optimization		Quantum advantage
Fast computation		Compact learning

Together they build powerful Quantum Machine Learning models!

Framework = Bridge Between Classical AI and Quantum Computing

Choose the right framework, the right optimizer, and build the future of intelligent quantum systems!

Series #53

Become a Quantum Machine Learning (QML) Engineer

Quantum Backpropagation Analogues

How Hybrid Quantum Neural Networks Learn

HARIKARAN M
AI Researcher

This is one of the most misunderstood topics in Quantum Machine Learning.

Many beginners ask:

"Does Quantum Machine Learning use backpropagation like Deep Learning?"

The answer is: Yes and No.

- ✅ Yes, hybrid QML models use backpropagation across the complete network.
- ❌ No, the quantum circuit itself does not use classical backpropagation internally.

⭐ Instead, quantum circuits use the Parameter-Shift Rule (or another quantum gradient method), while the classical layers use ordinary backpropagation.

LEARNING ROADMAP

Classical Backpropagation
↓
Why Quantum is Different
↓
Hybrid Backpropagation
↓
Chain Rule
↓
Gradient Flow
↓
PennyLane
↓
PyTorch Integration
↓
TensorFlow Integration
↓
Medical Imaging Example

1. WHAT IS BACKPROPAGATION?

In Deep Learning, backpropagation is the algorithm used to train neural networks.

It calculates how much each weight contributes to the error. Then weights are updated.

Classical Neural Network

Input → Dense Layer → ReLU → Dense Layer → Prediction → Loss → Backpropagation → Update Weights

Classical Formula

Suppose

Loss = L

Weights = W

Gradient

$$\frac{\partial W}{\partial L}$$

Optimizer
↓
Updates W

Example
Prediction = 0.80
True Label = 1.00
Loss = 0.20

2. WHY DOESN'T CLASSICAL BACKPROPAGATION WORK?

Suppose our circuit is

Input → RY → RX → CNOT → Measurement

The problem is Measurement.

Once we measure, the quantum state collapses. The internal amplitudes are no longer directly available for classical differentiation.

Therefore, ordinary backpropagation cannot simply "flow through" the quantum hardware.

The Solution

Instead of backpropagation inside the quantum circuit, we use Parameter-Shift Rule to compute quantum gradients.

Then, those gradients become part of the overall optimization process.

3. HYBRID BACKPROPAGATION

The complete network looks like

Input Image
↓
CNN
↓
Feature Vector
↓
Quantum Layer
↓
Measurement
↓
Dense Layer
↓
Prediction
↓
Loss

GRADIENT FLOW

Loss
↑
Dense Layer
↑
Quantum Layer
↑
CNN

Classical layers use classical gradients. Quantum layer uses quantum gradients. Together they form one optimization graph.

4. WHAT IS THE CHAIN RULE?

The chain rule connects gradients across multiple layers.

Image → CNN → Quantum Layer → Classifier → Loss

The total gradient is computed by combining the gradient from each stage.

Mathematically,

$$\frac{\partial x}{\partial L} = \frac{\partial y}{\partial L} \times \frac{\partial z}{\partial y} \times \frac{\partial x}{\partial z}$$

Gradient of Loss wrt Input — Gradient from next layer — Gradient from previous layer

Each layer contributes part of the gradient.

5. HYBRID GRADIENT FLOW

Suppose

CNN → 128 Features → Quantum Layer (4 Outputs) → Dense Layer → Loss

Gradients flow like

Loss
↓
Dense Layer Gradient
↓
Quantum Gradient (Parameter-Shift Rule)
↓
CNN Gradient

Notice the quantum gradient comes from the Parameter-Shift Rule instead of ordinary backpropagation.

6. COMPLETE TRAINING LOOP

Forward Pass → CNN → Quantum Circuit → Measurement → Prediction → Loss → Parameter-Shift Rule → Parameter-Shift Rule → Quantum Gradient → Optimizer → Update Parameters (θ) → Forward Pass

Repeat Until Convergence

7. WHAT HAPPENS INTERNALLY?

Input
↓
Quantum Layer
↓
Output

During training, PennyLane performs:

$\theta + \frac{\pi}{2}$ → Run Circuit
↓
$\theta - \frac{\pi}{2}$ → Run Circuit
↓
Compute Gradient
↓
Send Gradient to PyTorch

PyTorch continues normal backpropagation.

8. MEDICAL IMAGE EXAMPLE (HYBRID QML)

MRI Image
↓
EfficientNet-B0
↓
128 Features
↓
Linear Layer
↓
4 Features
↓
Angle Embedding
↓
4-Qubit PQC
↓
Measurement
↓
Dense Layer
↓
Softmax
↓
Prediction

TRAINING FLOW

CrossEntropy Loss
↓
Backpropagation
↓
Quantum Gradient (Parameter-Shift Rule)
↓
CNN Gradient
↓
Adam Optimizer
↓
Update All Parameters

Both classical and quantum parameters learn together.

9. WHICH PARAMETERS ARE UPDATED?

CLASSICAL PART

CNN Weights

Dense Layer Weights
↓
Updated by Classical Backpropagation

QUANTUM PART

$RY(\theta_1)$

$RX(\theta_2)$

$RZ(\theta_3)$

⋮

$RZ(\theta_n)$

Updated using Parameter-Shift Rule followed by optimizer.

10. CLASSICAL vs HYBRID TRAINING

	Classical AI	Hybrid QML
How it Learns	Backpropagation everywhere	Backpropagation + Parameter-Shift
Parameters	Neural weights	Neural weights + Quantum parameters
Graph Type	One computation graph	Mixed classical + quantum graph
Gradients	Gradients from calculus	Classical gradients + quantum gradients
Optimizers	Adam / SGD	Adam, SPSA, COBYLA, etc.

11. ADVANTAGES & LIMITATIONS

ADVANTAGES

- ✅ End-to-end learning
- ✅ Works with PyTorch and TensorFlow
- ✅ Automatic differentiation
- ✅ Supports hybrid CNN + PQC models
- ✅ Enables training of Quantum Neural Networks (QNNs)

LIMITATIONS

- ❌ Quantum gradients require many circuit evaluations.
- ❌ Training becomes slower as the number of quantum parameters increases.
- ❌ Noise on real hardware can affect gradient quality.
- ❌ Deep PQCs may still suffer from barren plateaus.

12. POPULAR FRAMEWORKS & THEIR ROLE

PennyLane	Hybrid QML framework Automatic gradients via Parameter-Shift Rule Works with PyTorch & TensorFlow	Best for Hybrid QML
Qiskit	IBM Quantum ecosystem Circuit design & execution Access to real quantum hardware	Best for IBM Quantum
TensorFlow Quantum (TFQ)	TensorFlow integration Quantum neural networks Hybrid deep learning models	Best for Deep Learning
Cirq	Google's quantum SDK Circuit construction & simulation NISQ research	Best for Research

13. INTERVIEW QUESTIONS

1. Does QML use backpropagation like Deep Learning?
2. Why can't we use backpropagation inside a quantum circuit?
3. How does PennyLane support hybrid training?
4. Which gradients are updated during hybrid training?
5. Which rule computes gradients for the quantum layer?

14. KEY TAKEAWAYS

- Classical backpropagation cannot flow through measured quantum circuits.
- ✅ Quantum gradients are computed using the Parameter-Shift Rule.
- ✅ Hybrid QML combines classical backpropagation with quantum gradients.
- ✅ Frameworks like PennyLane connect seamlessly with PyTorch & TensorFlow.
- ✅ This hybrid gradient flow is the foundation of modern Quantum Neural Networks.

ONE-LINE SUMMARY

Hybrid QML uses backpropagation across the network, while quantum layers compute gradients using the Parameter-Shift Rule — enabling end-to-end learning on today's quantum hardware.

Series #54

Become Quantum Machine Learning (QML) Engineer

Measurement Strategies in Quantum Machine Learning

The Bridge Between the Quantum World and the Classical World

HARIKARAN M
AI Researcher

This is the final and one of the most important topics in Module 3.
Without measurement, a quantum computer cannot provide useful information to a classical computer.
In Quantum Machine Learning (QML), measurement converts the quantum state into classical data, allowing us to compute predictions, losses, and gradients.

LEARNING ROADMAP

What is Quantum Measurement?
↓
Wavefunction Collapse
↓
Measurement Bases
↓
Expectation Values
↓
Sampling
↓
Probability Measurements
↓
Shots
↓
PennyLane
↓
Qiskit
↓
QML Applications

1. WHAT IS QUANTUM MEASUREMENT?

Simple Definition

A quantum measurement is the process of observing a quantum state and converting it into classical information.

Think of measurement as:

Quantum World → Measurement → Classical World

Real-Life Analogy

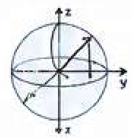
A sealed box containing a spinning coin.

Before opening:		After opening:
Head + Tail (similar to superposition)	→	Head or Tail (only one outcome appears) H T

Without measurement: We cannot use the quantum result in machine learning.

2. WHY DOES MEASUREMENT COLLAPSE THE QUANTUM STATE?

Before measurement:



Example

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This means
50% $|0\rangle$
50% $|1\rangle$

↓

After measurement:

$|0\rangle$ or $|1\rangle$

The superposition disappears.
This is called wavefunction collapse.

Why It Is Important?

A classical optimizer cannot understand a quantum state directly. Measurement converts it into numbers like:

0.73 | -0.42 | 0.91 | …

Now the optimizer can compute a loss.

3. MEASUREMENT BASES

A qubit can be measured in different bases.

Pauli-Z Measurement (Most Common)

Z
- Measures $|0\rangle$ or $|1\rangle$
- Eigenvalues returned: +1 or -1
- Almost all QML models use Pauli-Z expectation values.

Pauli-X Measurement

X
- Measures the qubit in the X basis.
- Useful in: quantum algorithms, communication, sensing.
- Less common in standard QML.

Pauli-Y Measurement

Y
- Measures in the Y basis.
- Useful for specific quantum physics applications.
- Rarely used in basic QML.

Comparison

Basis	Common in QML?	Purpose
Pauli-Z	★★★★★	Classification, regression
Pauli-X	★★	Special algorithms
Pauli-Y	★	Physics applications

4. EXPECTATION VALUE

Instead of measuring only one outcome, we estimate the average value.

Definition

The expectation value is the average measurement result obtained over many repeated executions of the same circuit.

Mathematically,

$$\langle Z\rangle = \langle\psi | Z | \psi\rangle$$

where $|\psi\rangle$ = quantum state, Z = Pauli-Z operator

Why Use Expectation Values?

- ✓ Stable
- ✓ Smooth
- ✓ Differentiable
- ✓ Excellent for optimization

PennyLane Example

```
return qml.expval(qml.PauliZ(0))
```

Meaning:

Measure qubit 0 and compute its average Pauli-Z value.

Example

Results over 5 shots:

+1 | +1 | -1 | +1 | +1

$$\text{Average} = \frac{(+1+1-1+1+1)}{5} = 0.60$$

Expectation value = 0.60

Expectation Value vs Sampling

Expectation Value	Sampling
Average result	Individual outcomes
Stable	Noisier
Differentiable	Less convenient for optimization
Most common in QML	For statistics & some algorithms

5. WHAT ARE SHOTS?

A quantum computer repeats a circuit many times.
Each repetition is called a shot.

Example: 100 shots

Run circuit 100 times → Estimate results

Why Are Shots Needed?

Quantum measurements are probabilistic.

Example (100 shots)

$|0\rangle$: 55
$|1\rangle$: 45

Probability:
$P(|0\rangle) = 0.55$ $P(|1\rangle) = 0.45$

More Shots = Better Estimates

Shots	Accuracy	Runtime
10	Low	Very Fast
100	Medium	Fast
1000	High	Slower
10000	Very High	Much Slower

Simulator

Infinite shots
Infinite shots (Analytical expectation values)

Real Hardware

Finite shots (1024, 4096, 8192)
Noise present

6. MEASUREMENT IN HYBRID QML

MRI Image
↓
EfficientNet
↓
Feature Reduction
↓
4-Qubit PQC
↓
Measurement → $\begin{bmatrix} 0.71, & -0.20, \\ 0.84, & 0.31 \end{bmatrix}$
↓
Dense Layer
↓
Prediction

Measured values are classical features that become input to the next classical layer.

7. WHICH MEASUREMENT STRATEGY SHOULD YOU USE?

Task	Best Choice
Image Classification	Expectation Values ★
Regression	Expectation Values ★
Quantum State Analysis	Probability
Algorithm Debugging	Sampling
Hardware Characterization	Sampling + Probabilities

8. BEST PRACTICE FOR MEDICAL IMAGE CLASSIFICATION

MRI
EfficientNet-B0
128 Features
4 Features
4-Qubit PQC
Expectation Values (Pauli-Z)
Dense Layer
Softmax
Prediction

Use qml.expval(qml.PauliZ(i)) because expectation values are:

- Smooth
- Stable
- Compatible with gradient-based optimization
- Ideal for hybrid classifiers

9. WHICH PARAMETERS ARE UPDATED?

Classical Part

- CNN Weights
- Dense Layer Weights
- …

↓
↓

Updated by Classical Backpropagation

Quantum Part

RY(θ_1)
RX(θ_5)
RZ(θ_6)
⋮
↓

Updated using Parameter-Shift Rule followed by optimizer

10. CLASSICAL vs HYBRID TRAINING

	Classical AI	Hybrid QML
Training Method	Backpropagation everywhere	Backpropagation + Parameter-Shift
Parameters	Neural weights	Neural weights + Quantum parameters
Computation Graph	One computation graph	Mixed classical + quantum graph
Gradients	From calculus	Classical + quantum gradients
Advantages	Well established	End-to-end learning with quantum advantage

11. ADVANTAGES

- End-to-end learning
- Works with PyTorch & TensorFlow
- Automatic differentiation
- Supports hybrid CNN + PQC models
- Enables Quantum Neural Networks (QNNs)

12. LIMITATIONS

- ✗ Quantum gradients require many circuit evaluations.
- ✗ Training becomes slower as the number of quantum parameters increases.
- ✗ Noise on real hardware affects gradient quality.
- ✗ Deep PQCs may suffer from barren plateaus.

13. INTERVIEW QUESTIONS

1. What is quantum measurement?
2. Why does measurement collapse the quantum state?
3. What is the most common measurement in QML?
4. Difference between sampling and expectation values?
5. What is a shot?
6. Why are expectation values preferred in QML?

14. KEY TAKEAWAYS

- Measurement converts quantum information into classical data.
- Measurement causes wavefunction collapse.
- Expectation values of Pauli-Z are most common.
- Sampling returns raw outcomes, expectation values return averages.
- Shots determine the quality of estimation.
- Measured outputs feed classical layers in hybrid QML models.

Measurement is the bridge between the quantum world and the classical world.
Without measurement, Quantum Machine Learning cannot learn, predict, or improve.